

Excavating grammatical fossils: Effects and causes in the semantic organization of Swahili noun classes

Matt Malone^{*1}, Jordan Kodner^{2,3}, and Spencer Caplan^{1,4}

¹Linguistics Program, CUNY Graduate Center

²Department of Linguistics, Stony Brook University

³Institute for Advanced Computational Science, Stony Brook University

⁴Cognitive Science Program, CUNY Graduate Center

April 29, 2026

Abstract

The extent of a semantic basis of Bantu languages’ noun class systems has been debated for decades, with proposals ranging from systematic feature-based assignment to total arbitrariness. Using distributional word embeddings trained on the largest annotated corpus of Swahili, we find that most noun classes are semantically “compact” above chance, suggesting at least partial organization by meaning. However, further analysis applying the Tolerance Principle, a theory of linguistic generalizations, finds just one of 18 previously proposed semantic features, HUMAN, is learnable as a productive rule. To explore the source of these otherwise unproductive patterns, we simulate earlier stages of Swahili through loanword elimination, finding that noun classes were both more compact and closer to productivity in earlier eras. Together, these distributional, acquisition, and diachronic results indicate that the semantic organization of Swahili noun classes is no longer a synchronic part of speakers’ grammars. We argue it is a grammatical fossil: a pattern once actively imposed by the grammar, now persisting only as an eroding remnant. More broadly, we propose a framework for identifying and interpreting grammatical fossils, drawing a crucial distinction between measurable patterns in language data and the grammatical mechanisms, or absence thereof, that generate them.

Keywords: Swahili Noun Classes, Grammatical Fossils, Distributional Semantics, Productivity, Language Change

Word count: 10,617

^{*}Corresponding author: mmalone@gradcenter.cuny.edu

1 Introduction

Linguistic analysis not only unearths patterns within language data, but also serves as evidence for the grammar that generates such data. While grammars necessarily control the patterns that surface in language data, signals in the output can arise for reasons beyond synchronic constraints of mental grammar. Put more simply, grammar must leave traces in usage, but patterns in usage need not be attributed to grammar—hence the abundance of both “accidental gaps” (e.g., [Frisch & Zawaydeh, 2001](#); [Gorman, 2013](#); [Halle, 1962](#)) and “accidental regularities” (e.g., [S. R. Anderson, 2016](#); [Blevins, 2004](#)) purported to exist across languages. A ripe candidate for such an accidental regularity is the semantic organization of Bantu noun class systems, whose underlying rules, as we will show, appear to only weakly enforced by the modern grammar, if at all.

Across Bantu languages with robust noun class systems, HUMAN words appear in what is conventionally¹ referred to as “Class 1/2.” Many trees are found in Class 3/4, fruits in 5/6, tools in 7/8, animals in 9/10, etc. Such trends have motivated proposals that Bantu grammars systematically assign words to classes based on their lexical semantics ([Contini-Morava, 1994](#); [Denny & Creider, 1986](#); [Zawawi, 1979](#)). A problem arises, however, when conducting a comprehensive, large-scale analysis that looks beyond the (small) set of words aligned to each noun class’ dominant semantic features. If nouns are causally organized by their meaning, we must wonder why Bantu languages such as Swahili, our Bantu language of focus, would place the word for ‘exam’ in the class for trees, ‘name’ in the class for fruits, and ‘potato’ in the class for tools. While any rule may have exceptions, the number of such exceptions in this particular case, as we demonstrate, is exceedingly large, and proposing a robust semantic structure that accounts for the class membership of each noun has proven difficult, if not impossible. The wide range of exceptions leads us to probe not just how, but whether, semantic rules currently govern these Bantu systems.

Work on the history of the Bantu family has argued that the noun-class system of Proto-Bantu featured more consistent semantic organization than its daughter languages (e.g., [Breedveld, 1995](#); [Demuth, 2000](#); [Givón, 1970](#)). Adopting this perspective, we hypothesize that the grammar actively determined noun class assignment semantically early in the prehistory of Swahili but no longer does. Until some point in time, semantic rules neatly forced trees into one class, tools into another, and so on; however, as these rules weakened, “exceptions” crept into the classes, making the vestigial patterns harder to recognize. The fact that many trees remain in Class 3 represents an accidental regularity of Swahili from a modern speaker’s perspective, in that we cannot attribute their shared membership to any synchronic rule of the grammar. In this case, speakers’ grammatical knowledge is not responsible for consistencies in the lexical output. We thus refer to the semantic organization of Swahili noun classes as a *grammatical fossil*, a pattern of language caused by historical rules no longer imposed or active in the minds of speakers. While others (e.g., [Campbell, 2013](#); [Greenough & Kittredge, 1901](#)) have used the term “fossil” to refer generally to outdated forms (e.g. the English verb suffix *-th*, as in the archaizing or Biblical ‘doeth’), we apply the term more narrowly to those grammatical patterns which permeate language data, despite their unproductive status. Like real fossils, grammatical fossils are remnants of previous eras, and like real fossils—which are no longer part of a living organism and thus gradually erode in the elements—grammatical fossils, which are no longer maintained by the grammar, also gradually erode as the language changes around them. This

¹[Meinhof \(1906\)](#), expanding on [Bleek \(1862\)](#), originated this numbered system for Bantu noun classes, which has remained the conventional notation ever since. The numbers refer to a reconstructed Proto-Bantu system, from which all contemporary Bantu languages descend, making the system amenable for typological and historical comparison. Odd-numbered classes refer to singular nouns with their plural counterpart the even-numbered class one higher. Because many singular-plural mappings are predictable, it is commonplace to refer to them as a pair.

framework proves particularly useful for analyzing Bantu noun class semantics, but could easily be applied to other linguistic fossils, as we discuss at the end of this paper.

Finding fossils requires going beyond identifying a pattern in language use (as in [Caplan & Djärv, 2019](#)). Were all regularities and gaps in data assumed to be consequences imposed by the grammar (as may be implicitly assumed), one might attribute English’s lack of **bnick* and *blick* to the same internal constraint ([Halle, 1962](#)). In reality, of course, the former is forbidden by phonotactic constraint while the latter is absent only incidentally. To confidently remark on Swahili semantics, our primary case study, we need go beyond a surface-level investigation.

This work employs three distinct methods that together paint a harmonious picture of Swahili noun classes’ grammatical status and supports the fossil hypothesis. Our investigation begins with a “big data” corpus analysis: we train a distributional word embedding model (word2vec) on the largest available annotated Swahili corpus in order to quantitatively measure the semantic compactness of each noun class pair. This computational approach reveals a degree of intra-class distributional compactness for most noun classes that is well above chance but is also well below manually selected groups of taxonomically coherent noun groups (e.g., a set of animals or vehicles). These results effectively put to rest any proposal for semantically arbitrary or hyper-dominant class assignment. Yet these results alone do not provide direct insight into the grammar. Thus, we follow this with a quantitative analysis of productivity to determine whether Swahili learners can, in fact, acquire mappings between semantic features and noun classes. Results from applying the Tolerance Principle ([Yang, 2016](#)) to a lexicon derived from the same corpus data suggest that speakers learn *only* that HUMAN nouns productively belong in Class 1/2, with all other 17 previously proposed feature-class pairings found to be unproductive. These facts together are suggestive that the semantic organization of Swahili noun classes is a fossil; but to confirm this, we probe the trajectory of semantic trends over time. If nearly all semantic constraints in the lexicon are indeed unproductive, we expect to see classes’ semantic compactness erode over time. We confirm this is the case by conducting a diachronic study that simulates and re-analyzes earlier stages of Swahili prior to loanword introduction of various kinds; finding that noun class coherence increases as loanwords are removed.

In what follows, we apply these three sets of analyses in sequence to establish the semantic organization of Swahili noun classes as a grammatical fossil: a pattern once imposed by the grammar but now in the process of eroding, leaving only historical traces in the absence of a synchronic rule. Section 2 addresses whether a semantic pattern truly exists among Swahili noun classes. Finding that it does, Section 3 probes whether that pattern reflects a synchronic rule (re)imposed on the language by learners. As this appears not to be the case, Section 4 investigates the source of these largely unproductive semantic patterns. Finally, in Section 5, we discuss potential limitations, highlight several other candidate fossils, and propose a general framework for distinguishing grammatical causes from historical effects in language.

2 Distributional Analysis

The first step in determining the grammatical status of a phenomenon is to clearly demonstrate its effects above expected variation in the language. In other words, a pattern must be convincingly shown to exist before positing anything about a rule that generates it. Our first analysis therefore addresses this question: to what extent are noun classes in Swahili organized semantically? If the answer is “to no extent,” as [Richardson \(1967\)](#) and others suggest, further investigation becomes moot.

While we are not the first to investigate the semantics of Swahili noun classes, we are, however, the

first to use a large (25M token) Swahili corpus to conduct an analysis of this nature at scale. Unlike prior research, our analysis does not select a subset of nouns in each class to investigate, or tag words according to their bare meanings (i.e., using a dictionary) rather than role in contemporary usage. Using a distributional model, we run calculations on embeddings for over 20k noun types, providing a comprehensive study of distributional “semantic compactness” for each noun class pair.

Like most Bantu languages, Swahili has a morphologically robust noun class system which categorizes nouns into groups, each marked by prefixation. The prefixes in Table 1 — displaying each of Swahili’s noun classes² with their prefixes along with an example word from each class — could be analogized to noun suffixes *-o* and *-a* in Spanish, which typically denote a masculine and feminine noun, respectively. The primary difference between such gender marking in Romance languages is that Bantu languages simply distinguish a larger number of classes, typically between 10 and 20. Note that classes are *paired* for the most part, with singular classes denoted by odd numbers and their plural counterparts denoted by the next highest even number. The numbers used to denote these classes follow the convention of studies in Bantu linguistics (Bleek, 1862; Meinhof, 1906), and are meant to represent cognate classes across Bantu languages. The numbers themselves are etymological, based on reconstructed Proto-Bantu, and are therefore not meaningful for synchronic purposes nor specific to Swahili. Meinhof (1906) reconstructed 23 classes, though no Bantu language today has that many classes. Languages described to have the closest number to 23 are Kagulu with 18 (Petzell, 2008) and Venda with 17 (Poulos, 1990).

Not all noun classes follow this neat singular-plural pairing. Swahili’s class 6, for instance, also contains mass nouns such as *ma-ji*, ‘water,’ which lack any singular counterpart. Class 9 nouns occasionally take a Class 6 prefix rather than Class 10 in the plural, e.g., *ma-rais*, ‘president(s).’ Most complex is Class 11, which results from a merger between historical Class 11 and 14 which both lost their consonantal prefixes (*lu-* and *wu-*), leading to syncretism between the forms. As a result, the singular class has various pluralization patterns: some nouns take a Class 6 prefix, others a Class 10, and others are abstract nouns with no plural form at all; e.g., *usahaulifu*, ‘forgetfulness’ (Contini-Morava, 2011).

Class	Prefix	Example	Class	Prefix	Example
1	<i>m-</i>	<i>m-toto</i> , ‘child’	2	<i>wa-</i>	<i>wa-toto</i> , ‘children’
3	<i>m-</i>	<i>m-ti</i> , ‘tree’	4	<i>mi-</i>	<i>mi-ti</i> , ‘trees’
5	$\emptyset$ - / <i>ji-</i>	<i>ji-cho</i> , ‘eye’	6	<i>ma-</i>	<i>ma-cho</i> , ‘eyes’
7	<i>ki-</i>	<i>ki-ti</i> , ‘chair’	8	<i>vi-</i>	<i>vi-ti</i> , ‘chairs’
9	$\emptyset$ -	<i>nyumba</i> , ‘house’	10	$\emptyset$ -	<i>nyumba</i> , ‘houses’
11	<i>u-</i>	<i>u-pendo</i> , ‘love’			

Table 1: Swahili noun class prefixes and representative examples. Numbering here follows the Bleek-Meinhof system (Bleek, 1862; Meinhof, 1906) used to characterize Proto-Bantu, wherein singular and plural forms are split into odd and even pairings. Contemporary Bantu languages have maintained some but not all of the ~ 20 reconstructed original classes. When analyzing synchronic Swahili, the numbers on their own carry no meaning.

An important distinction ought to be made here between noun class and agreement pattern. Each noun class triggers a distinct agreement pattern which surfaces on modifying elements like adjectives, numerals, and demonstratives. These patterns can distinguish classes whose noun prefixes are identical. For example,

²Omitted from Table 1 are Classes 15-18, as they are not related to semantic properties of nouns and thus, outside the current scope. Class 15 contains all verbal nouns (e.g. *ku-kimbia*, ‘running’), and Classes 16-18 contain locative constructions (e.g. *nyumba-ni*, ‘at the house’). Thus, these classes behave differently from Classes 1-14, in that they are taken to reflect derivational morphology.

Class 1 and Class 3 share a noun class prefix (*m-*), but agreeing elements like demonstratives distinguish them: *huyu* vs. *huu*. In rare cases, a noun will use agreement patterns of a class that does not match their noun prefix. An important instance of this case is discussed in Section 3 in the context of HUMAN nouns outside of Class 1/2. Following Carstens (2005), we take agreement to operate through the Minimalist Agree operation (Chomsky, 2000, 2001), wherein uninterpretable ϕ -features on nominal heads are valued by higher functional probes, driving agreement on targets such as verbs, adjectives, and demonstratives. Nevertheless our focus here is on the assignment of nouns to their various classes rather than the agreement such classes trigger.

The degree to which Swahili (and more generally, Bantu) noun classes are semantically motivated, if at all, has been debated since the mid-20th century. Early claims assert a lack of semantic motivation to Bantu noun class systems. Soon after providing the most robust Bantu classification typology, Guthrie (1948:11) claims that “no correlation of the [Bantu] genders with sex reference or with any other clearly defined idea.” In line with this belief, Richardson (1967:378) state this position quite strongly, “...it is impossible to prove conclusively by any reputable methodology that nominal classification in Proto-Bantu was indeed widely based on conceptual implication...”

In response, and in contrast, subsequent work attempted to clearly outline the semantic organization of Proto-Bantu and its descendants. Denny and Creider (1986), for one, associate each Proto-Bantu class with a unique set of physical characteristics, such as “solid non-extended units” for Class 5. Their taxonomy of physical properties for count nouns is provided in Figure 1A. Zawawi (1979) offers a similar breakdown for Swahili specifically, but argues that lines should be drawn between nominal prefixes rather than agreement patterns. For example, *m(w/u)-*, the nominal prefix for both Class 1 and Class 3 nouns all correlate to “substances of life” Zawawi (1979:72). A more detailed semantic breakdown is offered by Contini-Morava (1994), which constructs “networks” of connected major and minor semantic properties for various Swahili noun classes. Figure 1B shows Contini-Morava’s (1994:9) network constructed for Class 3 nouns.

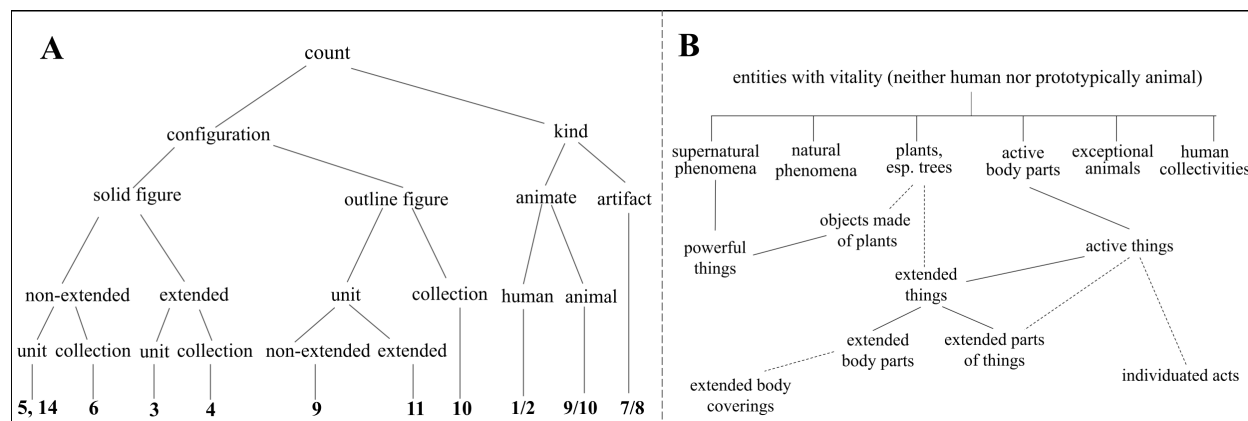


Figure 1: Left (A): an adapted diagram of Creider (1975)’s Semantic Analysis of Proto-Bantu Classes. The numbers as terminal nodes represent Bantu noun classes. Right (B): adapted diagram of “A semantic network for Class 3” from Contini-Morava (1994). Higher categories are meant to be more general, lower more specific. Following Langacker (1988), solid lines are meant to represent “relations of schematicity” and dotted lines are meant to represent “relations of extension.”

Each of the above exemplifies a worthy effort to justify and characterize the semantic nature of Bantu or Swahili noun classes, but the work has been limited to small, selective datasets that make use of words

out of context. [Contini-Morava \(1994\)](#) employs the largest dataset, a list of 4,000 Swahili words manually annotated with semantic features, while other experiments analyze as few as 100 words. The lexicons under analysis do not sufficiently represent the scope of nominals in Swahili, or any Bantu language.

We are not the first to note this specific critique. As early as the 1950s, while responding to [Guthrie’s \(1948\)](#) claim that noun classes are not semantically motivated, [Worsley \(1954:286\)](#) calls out the problem of data sparsity: “In spite of the undoubted correctness of his examples, it can be said that they do not disprove this correlation. The whole issue is obscured by the method of casual exampling which is adopted, although strictly this is not a methodical approach at all. If one applies quantitative analysis to the genders, one finds certain quite clear ideas contained in most genders.” Similarly and more recently, [Msaka \(2019:4\)](#) critiques the aforementioned Bleek-Meinhof system ([Bleek, 1862; Meinhof, 1906](#)) for characterizing the morphology and semantics of noun classes based only on “a partial dataset.” The irony is that both of these criticisms are followed themselves by quite limited analyses: [Worsley’s 1954](#) largest dataset consists of 133 words, and [Msaka \(2019\)](#) employs a monolingual Chichewa dictionary, analyzing words based on their definitions rather than use in context.

Beyond the issue of “casual exampling” or “partial datasets,” the methods employed to semantically characterize each noun class have been largely opaque. For example, [Contini-Morava \(1994\)](#) does not provide specifics on how many nouns in a given class need to be associated with a semantic feature before it is included in the network for that class. Using an example from Figure 1B, we wonder whether there exists some “lower bound” on the percentage of Class 3 words labeled POWERFUL to merit a “powerful things” node in the class’ semantic network, and how such a bound would be determined. Based on the methodology that is detailed, it is possible only a handful of nouns require a shared semantic feature to be associated with the class.

Computational approaches to the debate of noun class semantics, while admirable in their attempt to analyze the lexicon at scale, have their own problems. [Olstad \(2012\)](#) uses Bayesian inference to train a model that predicts the noun class of a Swahili word (based on the semantic features collected by [Contini-Morava](#)). [Olstad \(2012:180\)](#) finds that “Swahili noun class is not fully predictable by the model, but the model is particularly strong in predicting the ‘human’ class 1/2.” While approaching the type of work we find missing in the field, this supervised approach relies on the manual annotation of previous research as well as the limitations of [Contini-Morava’s \(1994\)](#) dictionary dataset. Moreover, a purely statistical analysis does not provide insight into any learning mechanisms; a model’s ability to predict feature-class mappings does not tell us anything about a human learner’s ability to generalize. Since statistical models can only tell us about the presence or absence of a pattern, not whether a learner could commit a pattern into the grammar, they cannot actually tell us whether the pattern is synchronically enforced or not.

More recently, [Palmieri and Kogkalidis \(2024\)](#) leverage word embeddings to investigate the distributional semantics of Swahili noun classes; however, to avoid the bias of agreeing elements³, they translate Swahili text into English prior to conducting any analysis. As they specify themselves, “mediating meaning through a foreign language carries the risk of inducing translation shifts and introducing inaccuracies” ([Palmieri and Kogkalidis, 2024:2](#)). This is particularly true of translation between English and Swahili, given their genetic distance and misaligned grammatical categories ([Vulić et al., 2020](#)). Thus, the analysis becomes less reliable as it has been pivoted through a wholly unrelated language.

³Each Swahili noun class has a (mostly) unique agreement prefix for modifiers like numerals, demonstratives, etc. Embeddings for nouns in the same class will therefore cluster in any distributional model if trained on an uncleaned corpus of natural speech. See Section 2.1 for details on avoiding this issue. In short, we lemmatize all tokens that agree morphologically with head nouns.

Given these limitations to prior work on noun classification, we first seek to investigate Bantu noun class semantics at scale. We employ word2vec (Mikolov, Chen, Corrado, & Dean, 2013), a distributional word embedding model, trained on the largest available annotated corpus of Swahili, to analyze whether noun classes in Swahili are truly semantically compact. This is an important baseline analysis, which establishes whether the classes have the potential to be semantically coherent enough to warrant further analysis.

2.1 Swahili Data and Computational Methods

To reflect a comprehensive sample of Swahili, word embeddings were trained on the largest available annotated corpus of Swahili, the Helsinki Corpus of Swahili 2.0 (25M tokens), a genre-agnostic collection of books, news articles, and Tanzanian parliamentary documents recorded between 1952 and 2015 (Hurskainen, 2016). Table 2 provides an overview of the nouns in this corpus, including token/type counts broken down by noun class. With 20,400 distinct nouns in our evaluation set, our Bantu noun class analysis is, to our knowledge, the most comprehensive to date.

Morphological Gloss	Pair	Token Count	Type Count	Pair Tokens	Pair Types
1/2-SG	1/2	283,943	1,369	776,385	2,613
1/2-PL		492,442	1,244		
3/4-SG	3/4	423,927	1,133	605,914	1,817
3/4-PL		181,987	684		
5/6-SG	5/6	454,348	1,473	933,855	2,846
5/6-PL		479,507	1,373		
6-PLSG	6	183,320	223	188,916	319
6-PL		5,596	96		
7/8-SG	7/8	268,377	1,089	471,757	1,935
7/8-PL		203,380	846		
9/6-SG	9/6	225,692	448	281,764	789
9/6-PL		56,072	341		
9/10-SG	9/10	1,723,768	4,929	2,302,825	7,636
9/10-PL		579,057	2,707		
11/10-SG	11/10	139,123	178	159,798	320
11/10-PL		20,675	142		
11-SG	11	438,099	2,125	438,099	2,125
Total		6,159,313	20,400	6,159,313	20,400

Table 2: Token and type counts for the nouns evaluated in the semantic compactness analysis, drawn from the Annotated Helsinki Corpus of Swahili 2.0 (Hurskainen, 2016) and categorized by noun class. Token counts reflect total corpus occurrences; type counts reflect distinct noun forms. Nouns are grouped both individually (by singular/plural form) as well as by paired noun class. Mass nouns in Class 6 often have no plural, Class 9 nouns either pluralize with Class 6 or 10 morphology, and Class 11 nouns often lack a plural entirely.

The corpus required only limited pre-processing. Because distributional models like word2vec rely on word co-occurrence statistics, noun modifiers such as adjectives and numbers, which inflect according to the class of their head noun, had to be lemmatized prior to training embeddings. The reason is as follows:

Distributional models determine embeddings for noun types based on the words with which they co-occur most frequently. As a result, any distributional model will generate nearby embeddings for words like *cold* and *frigid*, as the two will each co-occur with words like *winter*, *temperature*, and *ice*. Because words of the same Swahili class co-occur with modifiers (e.g. demonstratives, adjectives, numerals) inflected to uniquely agree with their head noun, a model trained on an un-lemmatized corpus will naturally cluster words of the same noun class. Faced with this problem, [Palmieri and Kogkalidis \(2024\)](#) opted to translate their corpus into English because it lacks overt noun-modifier agreement. Instead, we opt to lemmatize inflected modifiers, making them “class neutral” and avoiding biased clustering.

Our word2vec model output a 300-dimension vector for each word encountered at least three times in the corpus. The quality of the resulting vectors was evaluated using Multi-Simlex ([Vulić et al., 2020](#)), receiving a 0.438 Spearman ρ correlation with human similarity judgments. [Vulić et al. \(2020:26\)](#) test various word embedding models (without unsupervised post-processing) against their gold standard and the highest correlation they find is $\rho = 0.439$. Other models like fastText and Bert-M perform as low as 0.332 and 0.125, respectively. Because our ρ value nearly matches this outcome, we consider the model well-suited for the present analysis.

We opted to use word2vec ([Mikolov et al., 2013](#)) as our embedding model not only because its vectors have been shown to accurately model lexical semantic judgments ([Hill, Reichart, & Korhonen, 2015](#)), but because unlike other models (e.g., fastText), word2vec is not sensitive to word-internal substring co-occurrences.⁴ Because most noun classes are marked by overt prefixes, substring sensitivity causes these groups to cluster in the embedding space based on their shared phonological forms. The goal of our study is to investigate purely semantic properties of the noun classes.

The “semantic compactness” of each noun class was measured by grouping vectors by class and calculating two metrics: the *average pairwise distance* of all vectors in the class, and the *average distance from the centroid* of each vector group ([Gómez-Adorno et al., 2017](#)).⁵ Because cosine distance correlates well with judgments of semantic similarity ([Bojanowski, Grave, Joulin, & Mikolov, 2017](#)), we can think of these compactness measures as a group of words’ internal similarity score. A group of vectors corresponding to animals, for example, should receive high compactness scores, while a group of randomly selected embeddings should receive a low score.⁶

These measures are useful in determining the *relative* semantic compactness of classes, but equally important are appropriate “ceiling” and “floor” compactness measures, between which the noun class scores

⁴To confirm that word2vec was the most suitable model to employ, a cursory experiment was conducted using fastText, which gave low compactness scores only to those classes without a nominal prefix, indicating that morphophonology was skewing results, as anticipated. The results of this experiment are included in Appendix A.3.

⁵Average pairwise distance is defined in Equation 1 and average distance from the centroid is defined in Equation 2, where x is one of n vectors in the set, c is the centroid, and d is the distance metric.

$$\text{Average pairwise distance} = \frac{1}{2n^2} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2 \quad (1)$$

$$\text{Average distance from centroid} = \frac{1}{n} \sum_{i=1}^n d(x_i, c) \quad (2)$$

⁶It should be noted that we found no systematic relationship between the number of items in a set and their embedding compactness measure — see Appendix Figure A.2. Were such a relationship to exist (e.g. if larger groups inherently showed lower compactness scores), it would be necessary to fix the set size of embeddings. Because this relationship does not exist, we confidently extracted the maximum number of noun types from each class in the corpus.

can be situated. Because a ceiling value should represent a group of closely related words, we take two such compactness measures for 110 animal words, and for 30 vehicle words. We selected these groups to compare noun classes to both a set of natural nouns and a set of artifact nouns, lest one group’s semantic properties be inflated or deflated by the embedding model.

To identify a reasonable lower bound — that is, the expected semantic compactness for entirely unrelated words — we select a group of 2,000 nouns⁷ from the corpus at random, irrespective of their noun class membership, and measure this group’s compactness. Because the compactness value depends on which nouns are randomly selected, we averaged and calculated the standard deviation of the compactness scores across 1,000 different random groups.

2.2 Semantic Compactness Results

Compactness scores for each Swahili noun class, as well as the “ceiling” groups of animals and vehicles, are provided in Figure 2 — here reporting compactness as measured by average pairwise distance, though we obtain qualitatively parallel results using a “distance from centroid” measure (see Appendix A.5). Each bar shows the difference between the compactness of the floor measure (words selected at random) and the noun group in question. The values thus indicate how much *more* or *less* compact than random a group is.

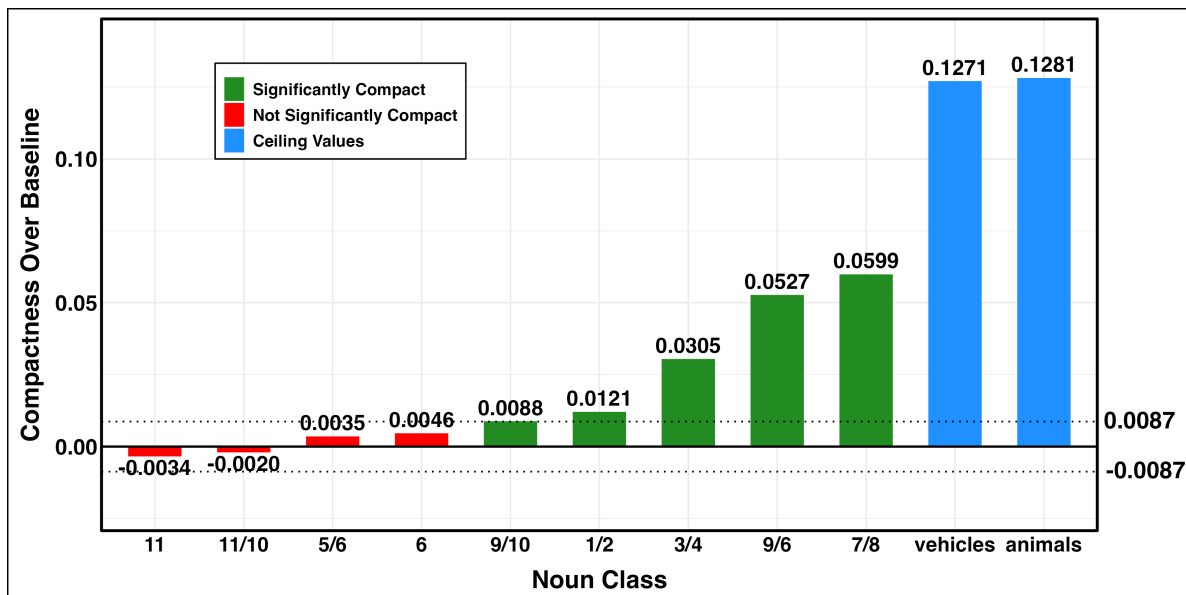


Figure 2: Semantic compactness of noun class groups alongside two manually constructed groups: animals and vehicles. The y -axis measures how much more compact these sets are than the average compactness of 1000 random sets (fixed at $y = 0$), with respect to vectors’ average pairwise distance. The higher the value, the more compact the sets were. The dotted lines represent two standard deviations above/below the random set’s compactness distribution. Red bars indicate values within two deviations, green bars are values above two deviations, and blue bars are ceiling values for comparison.

Five out of nine classes have compactness scores higher than two standard deviations of the random set’s distribution, while a handful of classes sit below this value. Classes 3/4 (typically associated with trees

⁷2,000 is roughly the average size of each set of noun class types though, as above, there is no systematic relation between set size and compactness score.

and plants), 7/8 (tools, manmade and small objects), and 9/6 (animals, some inanimate objects) are the most cohesive noun sets across both measurements of semantic compactness, with 1/2 and 9/10 close behind. Classes 5/6 (round things, fruits, liquids), 6 (mass nouns) and 11/(10) (abstract concepts, narrow things) have compactness scores below this threshold of two standard deviations. While the general contours of these scores are more important than each class’ compactness, it should be noted here that a likely reason for the particularly low compactness for 11 and 11/10 is the fact that Class 11, as discussed, is the result of two classes being merged, historically. If the original classes had distinct semantic motivations, their merger would have muddled them.

Were there no semantic basis to Bantu noun class assignment at all (as suggested by [Richardson \(1967\)](#)), we would expect compactness measures across all classes to sit near chance. Our data, however, shows most classes as notably more semantically compact than a random baseline, suggesting that semantic motivation is (or at some point, *was*) at play. Thus the answer to the first question at hand — whether a semantic pattern truly exists among items in Swahili noun classes — appears to be yes, though not an overwhelmingly strong one.⁸ These results alone, however, cannot adjudicate between an active grammatical rule and historical residue. While the fact that embeddings cluster together within noun classes serves as evidence, at least from the perspective of language use, of partial organization by meaning, a different approach is required to determine whether this pattern of usage reflects a psychological property, or merely the vestige of a former one.

3 Productivity Analysis

Evidence from distributional semantics (Section 2) suggests that Swahili noun classes are indeed semantically coherent to a degree, but this on its own is not enough to establish the synchronic status in the language, since it cannot disambiguate why any observed distributional patterns are present. While it is possible that the classes maintain their coherence due to the ongoing enforcement in the grammars of Swahili speakers, it is also possible that the classes are merely a remnant of a once productive pattern now lost in the language.

To answer this question, we must look beyond surface-level distributional cues and probe the question of Swahili grammar directly. To this end, we calculate the synchronic productivity of the noun classes with the Tolerance Principle (TP; [Yang, 2005, 2016](#)). We summarize our findings here: only HUMAN reliably predicts agreement with 1/2, in line with a range of previous findings ([Berghoff, Stockall, & Jonas, 2026](#); [Kanampiu, Martin, & Culbertson, 2025](#); [Olstad, 2012](#); [Palmieri & Kogkalidis, 2024](#)), while in the other direction, only Class 1/2, 11, and 11/10 productively predict the semantics of their members. These outcomes may appear counterintuitive in light of traditional studies which define classes semantically ([Denny & Creider, 1986](#)), in addition to our observed distributional compactness within many of the classes.

The TP is a cognitive model of generalization learning and productivity during language acquisition. When applied to concrete data, it serves as a quantitative metric for determining whether some linguistic

⁸These results also cast prior, small-scale analyses in a new light. [Denny and Creider’s \(1986\)](#) analysis of Class 3 nouns, for instance, draws on just 33 words from [Guthrie’s \(1971\)](#) Proto-Bantu reconstruction. These 33 nouns receive a compactness score of 1.249 — over 10 standard deviations more compact than the full 1,817 Class 3 nouns we analyze in the Helsinki Corpus of Swahili. To illustrate just how dramatically subset selection can distort results: examining all 50-word subsets of Class 3 nouns in the Helsinki Corpus, the maximum compactness score (1.264) is seven times, or over 123 standard deviations more compact than the minimum (0.193). Different subsets from the same data, even at a non-trivial size of $N = 50$, can be made to suggest that a noun class is either semantically incoherent or tightly organized. Without a principled, comprehensive sampling strategy, any conclusion about noun class semantics is liable to reflect the researcher’s choices as much as the data itself.

pattern is productive in a speaker’s grammar. For our purposes, this can be used to determine the synchronic reality of the Swahili noun classes. The TP was originally developed with morphology in mind, but it is independent of any particular level of the grammar. It has been applied to a wide range of phenomena across phonology, morphology, syntax, and semantics across English (Belth, Payne, Beser, Kodner, & Yang, 2021; Irani, 2019; Pearl & Sprouse, 2018; Richter, 2021; Yang, 2016), German (Belth et al., 2021; Yang, 2016), Icelandic (Björnsdóttir, 2021; Nowenstein, Sigurjónsdóttir, Yang, Ingason, & Wallenberg, 2020), Turkish (Belth, 2024), and Cree (Henke, 2023), among others, including the Bantu languages Chichewa (Msaka, 2019) and Kĩtharaka (Kanampiu et al., 2025). Motivated by research on pattern learning in psycholinguistics, the TP has been validated in a range of experimental studies spanning morphophonology, syntax, semantics and even domain-independent generalization of social conventions (Guilbeault, Caplan, & Yang, 2026; Koulaguina & Shi, 2019; Li & Schuler, 2023; Schuler, 2017; Schuler, Yang, & Newport, 2016; Shi & Emond, 2023).

Productivity, as conceived of under the TP, is a property of each individual’s grammar. It is crucially categorical: patterns, rules, or generalizations are either productive or they are not. This binary distinction has been conceptualized in many theories of grammar (J. M. Anderson, 1973; Aronoff, 1976; Kiparsky, 1973, among others) in terms of elsewhere conditions requiring irregulars to be looked up or accessed in some way before defaulting to a regular pattern. Since (unproductive) exceptions are ubiquitous across language, learners must be able to acquire, and speakers must be able to represent, patterns despite these exceptions. The TP defines the number of exceptions that a hypothesized generalization may tolerate beyond which the speaker must retreat to other generalizations or to memorization instead. The full cognitive motivation and mathematical derivation for the TP are beyond the scope of this introductory discussion, but see Yang (2005), Yang (2016, pp. 10, 144) for more information. The formal definition of the Tolerance Principle is given in Eq. 3:

$$e \leq \theta_N \quad \text{where} \quad \theta_N := \frac{N}{\ln N} \quad (3)$$

The tolerance threshold θ_N is the number of exceptions below which a learner can acquire a rule or generalization. θ_N is a function of N , the number of items in a given speaker’s lexicon which fall within the scope of the generalization and should in principle be subject to it, and e is the number of exceptions known to the speaker. The threshold is defined as the number of known types that a generalization should apply to divided by the natural logarithm of the number of types.

The TP is equipped to answer our questions regarding the productivity of Swahili noun classes. For example, it can tell us whether enough members of Class 1/2 are HUMAN to establish this as the productive HUMAN class, or whether there are too many non-HUMAN class members for this generalization to be encoded. In order to apply the TP to Swahili, we first need an estimate of a child Swahili learner’s nominal lexicon. Standard practice is to estimate this from a corpus (Yang, 2016). Cross-linguistically, children learn and accurately produce most of their native language’s morphology before school age. Swahili learners have even been shown to begin producing morphologically complex forms before age two (Deen, 2005), though it takes them years longer to achieve high consistency. Across languages, studies report that learners internalize productive vocabularies of 500-1,000 word types by around age 3 (Bornstein et al., 2004; Fenson et al., 1994; Hart & Risley, 1995; Kern, 2003; Szagun, Steinbrink, Franik, & Stumper, 2006), while estimates of English learners’ receptive vocabularies may reach 6,000 by first grade (Segbers & Schroeder, 2017). This suggests that we should consider no more than a few thousand words in our analysis. However, since we are interested specifically in nouns, which make up only a fraction of a learner’s vocabulary across

languages (Bornstein et al., 2004), we can reasonably conclude that children acquire their native language’s morphology on the basis of somewhere between several hundred up to perhaps a couple thousand noun word types.

Ideally, this lexicon would be estimated from a corpus of child-directed speech (CDS), since this is domain appropriate. Unfortunately, the CHILDES database provides no Swahili CDS corpora, so we must substitute something else. The lack of available CDS corpora is a persistent problem for not only Swahili, but also the vast majority of the world’s languages. Fortunately, following Kodner (2019), non-CDS corpora can be safely substituted for CDS for the purpose of obtaining lexicon counts for calculating morphological productivity. This insight has been leveraged for several studies modeling language change with language acquisition, since no historical CDS is available for any such language (Björnsdóttir, 2021; Kodner, 2023; Trips & Rainsford, 2022; Yang, 2024). The intuition behind this, laid out in Kodner (2019), comes both from work on child language acquisition and from natural language processing. Trips and Rainsford (2022) provides additional validation.

We summarize this intuition here. First, there is consensus from an unusually broad range of research traditions in developmental linguistics that productive generalizations are acquired on the basis of type frequencies in the lexicon rather than token frequency or the presence of specific lexical items (Aronoff, 1976; Baayen, 1993; Bybee, 1985; Elman, 1998; MacWhinney, 1978; Pierrehumbert, 2003; Yang, 2016). Consequently, the specific content of particular words present in the lexicon (derived from CDS or otherwise) is not load-bearing so long as the estimated lexicon is of approximately the correct size and contains roughly the correct proportion of words expressing the pattern in question. In this regard, the TP is especially forgiving, since the exact number of exceptions does not matter, as long as it falls on the correct side of the threshold. Looking ahead to our results, very few of the generalizations lie near the threshold, so even some deviation in the number of exceptions derived from measurement error would have no impact on our conclusions.

Second, the relative age at which words are acquired strongly correlates with their token frequencies (Braginsky, Yurovsky, Marchman, & Frank, 2019; Goodman, Dale, & Li, 2008; Palermo & Eberhart, 1968; Swingley & Humphrey, 2018). So, we can estimate a learner’s vocabulary by sampling, directly or probabilistically, several hundred of the most frequent words from a corpus. Then, since words, regardless of corpus genre, tend to follow a Zipfian distribution (Chan, 2008; Jelinek, 1997; Zipf, 1936), this process of sampling high-frequency words has the effect of removing the long tail. By eliminating the mostly low-frequency document-specific vocabulary, lexical distributions tend to converge when only a child lexicon-sized number of high-frequency words is sampled (Kodner, 2019). Thus, we can take the high-frequency portion of the sample to approximate the lexicon of a “typical” child learner.

Following this methodology, we extract the 1,000 most frequent noun types from the same Helsinki Corpus of Swahili which we used in the word2vec analysis. This provides a reasonable estimate of the set of nouns in the early lexicon over which Swahili learners acquire productive generalizations about noun class membership. We manually annotate this lexicon with 18 semantic features typically associated with Bantu noun classes (Denny & Creider, 1986). We adopt this particular schema for feature-class pairing following Kanampiu et al. (2025), which utilizes the TP in their study of Kĩtharaka, another Bantu language. The feature-class pairing is provided in the leftmost two columns of Table 3. Because certain classes are associated with more than one semantic feature, we additionally analyzed four *sets* of features, the combination of all features associated with classes 3/4, 5/6, 7/8, and 11/(10). This covers the possibility that a set of semantic features may productively predict noun class membership (or vice-versa) even if no feature on its own does.

Using these annotations, we calculate productivity in two directions: whether a noun’s semantics productively predict its class membership, and whether a noun’s membership in a class productively predicts its semantics. These directions establish the *necessary* and *sufficient* conditions for the generalizations, respectively. The results of these calculations are summarized in Table 3.⁹

A key exception, denoted by the asterisks in the table’s second row of data, comes from the fact that HUMAN nouns with non-1/2 class prefixes typically trigger *agreement* with Class 1/2, surfacing on subject markers, demonstratives, adjectives, etc. (Carstens, 1991) This is a rare instance in Swahili where noun class and agreement class are not aligned — in other cases, a noun’s prefix reliably predicts the agreement pattern it will trigger. Due to this unique divergence, we separately analyze Class 1/2 by noun prefix and agreement pattern. Because the corpus only tags nouns by their class prefix, agreement tags for HUMANS and Class 1/2 nouns were added manually, based on the most common pattern triggered by the type in the corpus.

Strikingly, applying the TP yields very few productive generalizations. HUMAN has sufficiently few exceptions to yield productive Class 1/2 agreement, but none of the other 23 feature→class rules are productive. There are four productive generalizations in the opposite direction predicting semantic features given class membership. These are Class 1/2 (noun class and agreement) with the HUMAN feature, as well as Class 11 and ABSTRACT and Class 11 with the feature set of NARROW, WAVY, COHESIVE MASS, ABSTRACT. This leaves the relationship between Class 1/2 agreement and the HUMAN feature as the only necessary and sufficient productive pattern in the data. Comparing this to our word2vec study, Class 1/2 is one of the noun classes which is more than two standard deviations above baseline compactness, albeit only very slightly.

The Kanampiu et al. (2025) study of Kĩtharaka, a Bantu relative of Swahili with a similar noun class system, yields similar results. In their wug test experiment, where they ask speakers to apply noun prefixes to novel words based on corresponding images, Kanampiu et al. find that the only non-evaluative feature found to be conclusively productive across the nine tested was HUMAN. An additional experiment involving the productivity of morphophonology (i.e. noun class prefixes), led the authors to conclude that semantic features play a much weaker role than morphophonology in noun class assignment. These results are again corroborated by Berghoff et al. (2026), whose auditory experiment on isiXhosa (another Bantu language) finds that semantic violations on HUMAN stems are judged much worse than violations on other stems, suggesting that the “animacy rule” is active, while others are not in the same way.

Purely computational approaches to the question of Swahili noun classes also found the relationship between HUMAN and Class 1/2 to be the most salient by far; the model produced by Olstad (2012) (p. 180) is “particularly strong” at predicting Class 1/2 among HUMAN nouns, and Palmieri and Kogkalidis (2024) reports the highest *F1* score for predicting Class 1/2 based on a word’s definition. The next highest score is 30% lower (for predicting Class 11), and no other class-feature *F1* scores are above 0.5. The general lack of productivity of semantic features apart from HUMAN is consistent with our findings.

For the vast majority of noun classes and their associated semantic features, constraints do not appear to be actively imposed by the mental grammar, as the number of exceptions vastly exceeds the tolerance threshold for nearly every feature-class pairing. Apart from the HUMAN feature, there appear to be no unique semantic properties sufficiently aligned with a noun class to achieve productivity. Thus, the pattern identified

⁹One mathematical consequence of the Tolerance Principle is that the threshold for generalization is more permissive when *N* is small. Considering this possibility, we repeat our calculations on nouns derived from only the 500 most frequent types in the corpus. The results are nearly identical to those found in Table 3, and can be found in the Appendix (Table A.1). Since our overarching finding is a lack of productive generalizations, moving to a larger vocabulary size approximating adult lexicons will not yield additional productive generalizations.

Semantic Feature $\Rightarrow$	Class	N	e	Class $\Rightarrow$	Semantic Feature	N	e
HUMAN	1/2	181	43	1/2	HUMAN	141	3
HUMAN	1/2*	181	0	1/2	HUMAN*	184	3
TREE	3/4	2	0	3/4	TREE	78	76
DISPERSIVE MASS	3/4	5	5	3/4	DISPERSIVE MASS	78	78
EXTENDED	3/4	19	13	3/4	EXTENDED	78	72
SPREAD	3/4	7	7	3/4	SPREAD	78	78
COHESIVE MASS	5/6	12	9	5/6	COHESIVE MASS	119	116
COHESIVE MASS	11, 11/10	12	12	11, 11/10	COHESIVE MASS	134	134
AUGMENTATIVE	5/6	6	6	5/6	AUGMENTATIVE	119	119
PLANT PART	5/6	4	3	5/6	PLANT PART	119	118
FRUIT	5/6	2	1	5/6	FRUIT	119	118
ROUND	5/6	8	6	5/6	ROUND	119	117
ARTEFACT	7/8	19	12	7/8	ARTEFACT	70	63
PEJORATIVE	7/8	5	4	7/8	PEJORATIVE	70	69
PLANT	7/8	3	3	7/8	PLANT	70	70
MANNER	7/8	0	0	7/8	MANNER	70	70
ANIMAL	9/10	8	4	9/10	ANIMAL	457	453
NARROW	11, 11/10	16	15	11, 11/10	NARROW	134	134
WAVY	11, 11/10	2	2	11, 11/10	WAVY	134	134
ABSTRACT	11, 11/10	416	301	11, 11/10	ABSTRACT	134	19
TREE + DISPERSIVE MASS + EXTENDED + SPREAD	3/4	33	25	3/4	TREE + DISPERSIVE MASS + EXTENDED + SPREAD	78	72
COHESIVE MASS + AUGMENTATIVE + PLANT PART + FRUIT + ROUND	5/6	32	25	5/6	COHESIVE MASS + AUGMENTATIVE + PLANT PART + FRUIT + ROUND	119	112
ARTEFACT + PEJORATIVE + PLANT + MANNER	7/8	27	19	7/8	ARTEFACT + PEJORATIVE + PLANT + MANNER	70	62
NARROW + WAVY + COHESIVE MASS + ABSTRACT	11, 11/10	446	330	11, 11/10	NARROW + WAVY + COHESIVE MASS + ABSTRACT	134	18

Table 3: Tolerance Principle analysis of rules between proposed semantic features and noun classes. Left-side columns test the necessary condition (whether feature predicts class), where N is the number of nouns bearing the feature and e is the number of those nouns *not* assigned to that class. Right-side columns test the sufficient condition (whether class predicts semantics), where N is the number of nouns in the class and e is the number not expressing the listed feature. Green highlighted rows indicate a productive pattern under the TP ($e < \theta$). Following Yang (2016), small samples ($N < 5$) are excluded. Asterisks denote cases where agreement class diverges from noun class prefix: HUMAN nouns without a Class 1/2 prefix typically trigger Class 1/2 agreement regardless.

in Section 2, while salient, does not appear to be enforced by the grammar but rather is a grammatical fossil. If such patterns persist without grammatical enforcement, they are subject to disruption by any process — e.g., borrowing, semantic drift or other types of lexical replacement, sound change obscuring morphophonology such as the merger between Classes 11 and 14, morphological analogy, and so on — that introduces exceptions without the corrective pressure of a productive rule. We should therefore expect them to erode over time; a prediction we test directly in Section 4.

4 Diachronic Erosion

Results thus far tell a seemingly contradictory story. Distributional analysis finds that noun classes are semantically compact, at least to a degree — lexical organization that would be vanishingly unlikely to emerge by pure chance. Yet a TP analysis indicates that semantic features are overwhelmingly unproductive when it comes to class assignment. If such constraints are not actively imposed by speakers, we wonder how noun classes have come to be semantically organized at all. The ultimate origin of the system is extremely old, and thus not investigable by direct observation:¹⁰ the noun-class system of Proto-Bantu, spoken around 3000 B.C., is argued to have featured more consistent semantic organization than its daughter languages (e.g., [Breedveld, 1995](#); [Demuth, 2000](#); [Givón, 1970](#)), with the Bantu migration underway by 1000 B.C. ([Beleza, Gusmao, Amorim, Carracedo, & Salas, 2005](#); [Vansina, 1995](#)). One likely driver of this simplification and subsequent erosion is the introduction of loanwords which, as we will show, seem to have entered Swahili classes on morphophonological rather than semantic grounds — disrupting the earlier system.

If the semantics of this system have been gradually eroding over time, we would expect earlier states of Swahili to contain both productive assignment rules and more semantically compact noun classes than those represented in the modern Helsinki Corpus. To address our third and final question, the stability and erosion of semantic patterns, we simulate earlier versions of Swahili and recalculate compactness and productivity by class.

Swahili speakers have come into contact with speakers of many other languages, incorporating thousands of loanwords across the lexicon as a consequence of a long history of trade with South Asia and the Middle East, as well as Portuguese and British colonization ([Schadeberg, 2009](#)). We extracted two sets (detailed in Table 4) from the WOLD database ([Haspelmath & Tadmor, 2009](#)) and Wiktionary ([Wiktionary contributors, 2025](#)): a “pre-modern” set of Persian and Arabic loanwords likely introduced between the 10th and 15th centuries, and an “early modern” collection from Hindi, Gujarati, Kachchi (all three are Indic languages), Portuguese, and English, whose speakers made contact with East Africans mostly after 1500. Eliminating these loanwords provides a proxy for “earlier versions” of Swahili, simulating the lexicon as it may have looked roughly 500 and 1,000 years ago.

Of course, the addition of loanwords is not the *only* difference between Swahili today and 1000 years ago. Sound changes have merged classes (e.g. Class 11 and 14), class prefixes have been lost (e.g. Class 5 or Class 9), and other internal changes have obscured the grammar active in pre-modern times. The methodology employed in this diachronic analysis should not be taken as ignorance of these facts; rather, we select one clear difference between snapshots of Swahili to investigate in isolation, and examine its

¹⁰There are, as far as we know, no recorded instances of a language *gaining* a robust noun-class system. All modern languages with large noun-class systems seem to have inherited them from parent languages (Don Ringe, pc), making the question of their ultimate source even more difficult to contend with.

Language	Loanword Count	Time Period
Arabic/Persian	1,642	Pre-Modern
English	993	Early Modern
Gujarati/Kachchi	23	
Hindi	25	
Portuguese	38	

Table 4: Loanwords eliminated from the evaluation set by source language and time period, used to simulate earlier Swahili lexicons.

effects on the two measures conducted in the preceding sections.

Compactness measures were recalculated after removing the set of early modern loanwords, and again after removing pre-modern borrowings. These calculations are compared to 1,000 trials of dropping the same number of randomly-selected words from each class, in both time periods. The overall results of these diachronic reanalyses are provided in Figure 3A, while Figure 3B breaks this data down by noun class.

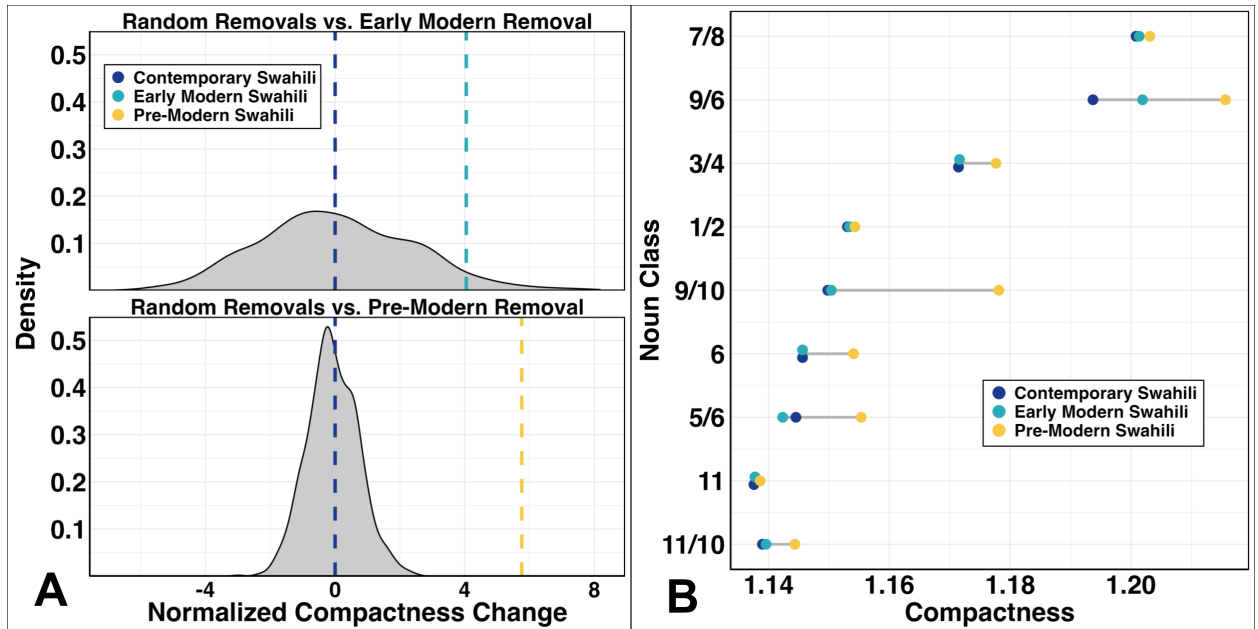


Figure 3: Left (A) shows two smoothed density histograms of the change in compactness (normalized by class size) after eliminating sets of randomly-selected words, with dashed vertical lines to represent the compactness change when eliminating *loanwords*. Centered at zero is the class compactness of contemporary Swahili. Right (B) compares the compactness of noun classes at each time period.

Figure 3A demonstrates the effect of dropping loanwords on noun classes' compactness change. Compared to random noun elimination, dropping loanwords has a much larger (positive) effect on compactness, indicating that “older” versions of Swahili had more semantically cohesive noun classes.¹¹ This is partic-

¹¹As discussed in Section 2, we detected no relationship between set size and compactness score, so there is no reason randomly eliminating words should consistently change compactness in a certain direction; however, we compare loanword elimination to

ularly clear in the case of pre-modern Swahili, whose average normalized compactness change (5.759) is 2.29 times larger than the highest normalized compactness change of the 1,000 random trials (2.520). The broader variance observed in the early modern condition is likely driven in part by the smaller proportion of nouns removed, which increases sensitivity to the particular items sampled in each trial. In contrast, larger removal sets in the pre-modern condition yield more stable outcomes, producing a sharply peaked distribution.

Figure 3B shows that the effect of loanword-dropping is not driven by one or a handful of classes; rather, *all* noun classes find highest compactness scores in the pre-modern era, with all loanwords eliminated. As for the early modern loanwords, only one noun class’ compactness did not increase when loanwords were eliminated. The outlier status of Class 5/6 may be due to the fact that many nouns for household items (e.g. *jagi* ‘jug’, *taulo* ‘towel’, *burashi* ‘brush’, *blanketi* ‘blanket’, *sinki* ‘sink’) were borrowed from English into this class in the early modern period. Because these loanwords occupy a similar semantic space, their introduction could contribute to making the class become *more* compact.

Both sub-plots in Figure 3 indicate that semantic compactness increases as loanwords (as opposed to randomly selected words) are eliminated from the evaluation set. These results are consistent with a hypothesis that the semantic organization of Swahili noun classes has been eroding across these periods of time. After loanwords are eliminated, not only do we find semantic compactness increasing, but *productivity* of more feature $\Rightarrow$ class pairings as well. As in Section 3, we test whether semantic features productively predict noun class, now on two additional sets of nouns: those lacking early modern loanwords, and then pre-modern loanwords. For each time period, the top 1,000 most frequent nouns are tagged by class and semantic feature, and productivity is calculated. Results are shown in Table 5.

Eliminating early modern loanwords shifts the balance of e and θ for the ANIMAL feature, such that the number of exceptions (animals *not* in Class 9/10) is below the tolerance threshold in this era. Eliminating pre-modern loanwords produces the same effect on the features DISPERSIVE MASS, ROUND, and HUMAN. As time moves forward, the number of productive feature-to-class rules decreases from 4 to 2 to 1. This effect of progressive tolerance is demonstrated in Figure 4, which visualizes the percent of e , compared to θ for each class.

While these results indicate clearly that noun class groups of an earlier era contained productive rules that are missing today, a notable question concerns how the language lost these productive rules initially. For example, DISPERSIVE MASS nouns were once productively assigned to Class 3/4 in Swahili, but today they are not. If language learners acquire and apply grammatical rules based on the input they receive from adults, an intervening force must have interrupted learners productively applying, for example, the rule mapping DISPERSIVE MASSES to Class 3/4.

A likely possibility relates to speakers’ tendency to reanalyze lexical borrowings. Amid the influx of Arabic words in Swahili, we see words like *mshahara* (from *mušāhara*, “monthly salary”). Salaries are not TREES or DISPERSIVE MASSES, nor are they physically EXTENDED or SPREAD—the four semantic features attributed to Class 3/4, the agreement class triggered by this loanword. Morphophonologically, however, a word beginning with *m-* is a good fit for Class 3, whose nouns already use this prefix. The loanword is re-segmented as a prefix (*m-*) plus a root (*shahara*), and then reanalyzed such that its plural form employs Class 4 morphology: *mi-shahara*.¹² Instances of this lexical reanalysis are not limited to a handful

dropping words randomly to demonstrate the significant effect on semantic cohesion that these borrowings hold.

¹²A more widely referenced example is Swahili’s adoption of the Classical Arabic *kitābu* ‘book’ as *kitabū*. In Arabic, this word is analyzable as a consonant root *KTB* occupying a particular vowel template with a nominative singular suffix *-u*. But in Swahili, the first sounds *ki-* are reanalyzed again as a prefix, rendering the remainder *tabu* as the root and spawning the plural *vi-tabu*, ‘books.’

Generalization		Pre-Modern		Early Modern		Contemporary (Table 3)	
Semantic Feature $\Rightarrow$	Noun Class	<i>N</i>	<i>e</i>	<i>N</i>	<i>e</i>	<i>N</i>	<i>e</i>
HUMAN	1/2	207	22	181	34	181	43
HUMAN	1/2*	207	0	181	0	181	0
TREE	3/4	2	0	2	0	2	0
DISPERSIVE MASS	3/4	7	3	6	5	5	5
EXTENDED	3/4	28	15	22	15	19	13
SPREAD	3/4	9	9	6	6	7	7
COHESIVE MASS	5/6	12	8	14	11	12	9
COHESIVE MASS	11, 11/10	12	11	14	13	12	12
AUGMENTATIVE	5/6	6	5	6	6	6	6
PLANT PART	5/6	11	6	7	4	4	3
FRUIT	5/6	2	1	2	1	2	1
ROUND	5/6	15	5	8	5	8	6
ARTEFACT	7/8	23	11	21	11	19	12
PEJORATIVE	7/8	4	3	5	4	5	4
PLANT	7/8	7	7	4	4	3	3
MANNER	7/8	0	0	0	0	0	0
ANIMAL	9/10	16	5	11	4	8	4
NARROW	11, 11/10	17	14	17	16	16	15
WAVY	11, 11/10	3	2	3	2	2	2
ABSTRACT	11, 11/10	370	192	427	299	416	301
TREE + DISPERSIVE MASS + EXTENDED + SPREAD	3/4	46	27	36	26	33	25
COHESIVE MASS + AUGMENTATIVE + PLANT PART + FRUIT + ROUND	5/6	46	25	37	27	32	25
ARTEFACT + PEJORATIVE + PLANT + MANNER	7/8	34	21	30	19	27	19
NARROW + COHESIVE MASS + WAVY + ABSTRACT	11, 11/10	402	219	461	330	446	330

Table 5: Tolerance Principle analysis of rules from proposed semantic features to noun classes, across three time periods: Pre-Modern, Early Modern, and Contemporary Swahili. Green highlighted cells indicate a productive pattern in the era. Following Yang (2016), small samples ($N < 5$) are excluded. The asterisk marks specify instances of agreement class rather than noun class (prefix). Each time period considers the top 1,000 most frequent types in the corpus, excluding any loanwords introduced after that period. Therefore, for a given feature, N may actually increase over time, as eliminating loanwords allows new types to enter the most frequent 1,000 types.

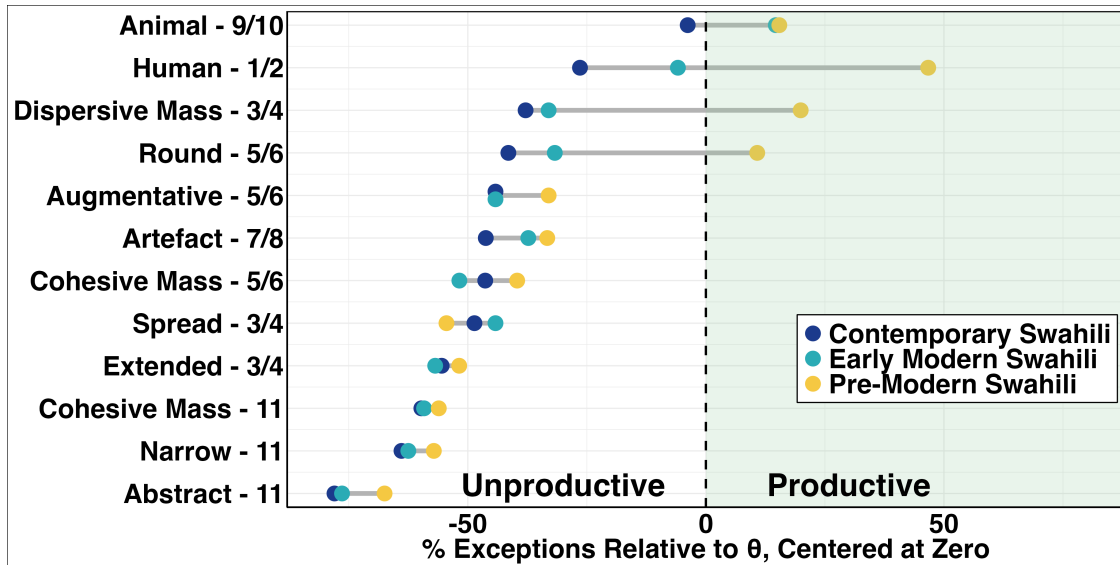


Figure 4: Productivity over time, by class-feature pairing. The dots measure the percent of exceptions that are above or below theta, for each class-feature pairing and time period in question. Green cells in Table 5 correspond to dots in the green area of this plot.

of loanwords. Of the top 130 most frequent loanwords in the Helsinki Corpus of Swahili¹³, 104 (81.3%) of the source words share an initial segment with the prefix of their assigned class. Examples include *ki-labu* (from the English ‘club’), pluralized *vi-labu*; *w-araka* (from the Arabic *waraqa*, ‘document’), pluralized *ny-araka*; *ch-epeo*, (from the Portuguese *chapéu*, ‘hat’), pluralized *vy-epeo*. Of the same 130 words, only 58 (45.3%) share semantic properties¹⁴ with the class to which they are assigned. Examples include *m-fereji* (from the Malagasy *farihy*, ‘canal’), pluralized *mi-fereji*, which shares the EXTENDED property of Class 3/4 nouns.

Note that many loanwords, such as *w-akati* (from the Arabic *waqt*, ‘time’), share *both* phonological and semantic properties with their assigned class.¹⁵ In such instances, it is not clear which of the two factors had stronger influence on class assignment. If we exclude such loanwords from the analysis, we are left with 76 words, 61 of which (80.3%) are motivated exclusively by phonological similarities to class prefix, and 15 of which (19.7%) appear to be integrated according to semantics alone.

Words whose morphophonology *does not* match an existing class prefix, e.g. the Arabic loanword *kahawa* (coffee), typically end up in Class 9/10, which lacks overt class prefixes. Were semantic properties central to loanword integration, we might expect *kahawa* to enter Class 6, which contains largely mass nouns and liquids.

Many have described this phonologically motivated process as central to loanword incorporation in Swahili (Gregersen, 1967; Polomé, 1967).¹⁶ It seems reasonable that speakers would have integrated enough

¹³Excluded here are the many words assigned to the prefix-lacking 5/6 and 9/10, which have become the default classes to which loanwords are assigned, based on no phonological or semantic basis. (Nurse, Hinnebusch, & Philipson, 1993). Also excluded are words that are *derived* from other loanwords, e.g. *u-misionari* ‘missionary work’, the abstracted form of *mmisionari*, ‘missionary.’

¹⁴Due to the smaller scale, this analysis employed both the semantic properties of Bantu languages assembled by Kanampiu et al. (2025) and the semantic webs of Swahili noun classes created by Contini-Morava (1994) were considered.

¹⁵Class 11 has the ABSTRACT property shared by the notion of ‘time’

¹⁶This discussion would be lacking without a mention of Zawawi (1979), which emphasizes the multiplicity of processes involved

words on this morphophonological basis to tip the number of semantic exceptions beyond the tolerance threshold, and decrease semantic coherence among classes. With enough loanwords disrupting these semantic regularities, productivity was lost and the rules began to erode. Together, the compactness and productivity results across simulated eras strongly favor the fossil hypothesis: what we observe in contemporary Swahili is the eroding residue of a once-active grammatical system, its traces now persisting in language use long after the grammar has ceased to enforce them.

5 General Discussion

Across the three sets of analyses conducted in this paper, a clear picture of Swahili noun class semantics emerges. Distributional analysis reveals that noun classes are semantically compact above chance, but nowhere near the coherence of taxonomically unified categories (e.g., animals or vehicles). A Tolerance Principle analysis then finds that this modest organization is overwhelmingly unproductive: of 18 previously proposed semantic feature-class pairings, only HUMAN reliably predicts Class 1/2 agreement, with all others falling well above the maximum threshold for tolerable exceptions. These two results together are already suggestive of a fossil, but neither alone is sufficient grounds to establish it. Distributional compactness without a test of productivity cannot distinguish an active grammatical rule from historical residue; a (negative) test of productivity without the distributional baseline would leave open whether any pattern existed to explain in the first place. They must be understood in tandem. Our diachronic analysis is further probative: by simulating earlier stages of Swahili through the elimination of loanwords, we find that both compactness and the progress towards productivity increases as we move back in time, confirming that what we observe in contemporary Swahili is the eroding residue of a once-active system. The convergence of all three results supports the conclusion that the semantic organization of Swahili noun classes is a grammatical fossil.

This description of Swahili may seem surprising given the confidence with which semantic rules for its noun classes have been proposed in the literature (Contini-Morava, 1994; Denny & Creider, 1986; Zawawi, 1979).¹⁷ But as we demonstrate in our distributional analysis, conclusions drawn from small or selectively assembled datasets are unreliable guides to latent structure. The famous case of English past tense offers a useful parallel: 54 of the 100 most frequent English verbs are irregular (Yang, 2016). If one were to attempt to construct a rule on that basis — as very young learners must — it would suggest irregularity is the norm. Yet among the 1000 most frequent verbs, only 13% are irregular, and the regular *-ed* pattern is robustly productive. Vastly different conclusions follow from systematically different samples. As in Section 2, a similar dynamic holds for Swahili noun classes, where compactness scores vary by well over 100 standard deviations across different 50-word subsets of the same noun class — a range that disappears only when the full lexicon is considered. Crucially, it is an individual’s lexicon, not any convenient subset, that constitutes the data from which learners must infer productive linguistic rules.

At the same time, this conclusion should not be taken to imply that “big data” analyses are sufficient on their own. No amount of language data can establish the existence of a grammatical rule without targeted tests of productivity. Put another way, the mere fact that a pattern is broadly observed in language use does

in assigning class to loanwords in Swahili, one of which is also phonology-based incorporation.

¹⁷Note that we do not deny that the agreement patterns triggered by noun classes (syntactic Agree operations) are synchronically active in the grammar, as discussed in Section 2, simply that no semantically-grounded rule productively assigns nouns to classes based on their meaning.

not entail a property of the grammar as its cause. The Swahili case illustrates this precisely. Several open questions remain, however, about the mechanisms by which this fossil came to be; and about what the fossil framework can and cannot tell us about the current state of Swahili grammar and Bantu noun class systems more broadly.

The first concern is the culprit of this non-productivity. What killed the productive semantic classification of (pre-)Swahili nouns in the first place and gave rise to this fossil, and why did it happen when it did? This is the Actuation Problem, the fundamental question of language change (Labov, Yaeger, & Steiner, 1972; Walken, 2017; Weinreich, Labov, & Herzog, 1968). The Actuation Problem is notoriously intransigent, because, by the time linguists have observed that a change has occurred, the moment of actuation is long past. It is rarely the case that the identity of the speakers who actuated the change is even recoverable, because early adopters and leaders of a change are typically not themselves the original innovators (Milroy & Milroy, 1985). This problem is of course magnified for changes which were actuated in the distant past or in times and places before mass literacy.

Our methods suggest a causal link between the borrowing, re-segmentation, and morphophonological sorting of loanwords and the death of productive semantic classification. The reasoning goes, following the logic of the Tolerance Principle, that morphophonologically sorted words are likely to constitute semantic exceptions for a given class, and when enough of these words enter the lexicon of a learner, the acquisition of productivity fails. 81.3% of Arabic loanwords which we observed were not sorted into Classes 5/6 or 9/10 were assigned for morphophonological reasons, as was the case for *mshahara* and *kitabu*. Plausible accounts for this type of actuation of changes to productivity via lexical change abound in the recent literature on the Tolerance Principle. Major changes to a system’s productivity may be the result of contact, as we propose for Swahili, or the result of unrelated phonological changes (Björnsdóttir, 2021; Dolatian & Kodner, 2026; Richter, 2021; Trips & Rainsford, 2022; Yang, 2024, *i.a.*).

A related question concerns the current synchronic basis of noun class assignment in Swahili, if not semantics. Our results establish what noun class assignment is *not* — it does not appear to be governed by productive semantic rules for the vast majority of classes — but the present results do not uniquely identify how native speakers in fact assign class membership to a novel form. The closest available evidence comes from Kĩtharaka, where Kanampiu et al. (2025) find that morphophonological properties of noun stems are a much stronger predictor of class assignment than semantic features, consistent with the loanword incorporation patterns we observe in Swahili. Whether morphophonology is similarly dominant across the full Swahili lexicon, and not just for loanwords, remains an open empirical question. It is worth noting, however, that this question may be less pressing in practice than it appears. Because noun class triggers robust agreement morphology on modifying adjectives, demonstratives, subject markers, and other predicates, a speaker encountering an unfamiliar word in naturalistic speech or text will typically be able to infer its class directly from the agreement morphology in the surrounding context — often before or independently of inferring its meaning. The situation faced by participants in wug-style experiments, who must assign a prefix to a novel form presented in isolation, is therefore somewhat artificial. Nevertheless it remains a valuable tool for isolating the cues speakers can draw on in the absence of contextual agreement information.

It further bears emphasis that the present study has focused on standard written Swahili specifically, owing to the relative abundance of annotated corpus data available for this language. This does not entail that the grammatical status of noun class semantics is uniform across Bantu languages or even among Swahili’s closest relatives. Swahili is, by most accounts, among the morphologically simpler Bantu systems, having undergone considerable simplification. This may be expected, given Swahili’s status as a regional lingua

franca. The effect is particularly strong in urban varieties, with Nairobi Swahili representing an extreme case: verbal subject agreement collapses all non-HUMAN noun classes to a single *i-/zi-* pattern, whose forms are reflexes of Class 9/10 among many Bantu languages, retaining distinct agreement morphology only for Class 1/2 animate nouns (Deen, 2002). Strikingly, this is precisely the one class whose semantic basis we find to be productive in standard Swahili. Considering the overt Class 9/10 marking in this system, the dramatically leveled Nairobi Swahili HUMAN/non-HUMAN pattern cannot have descended solely from Standard Swahili, and must have included some substantial influence from related varieties. Its actuation likely lies in the conditions of language contact in the diverse urban demographics of Nairobi. It is additionally possible that standard Swahili itself has been impacted in a similar way to a smaller degree, with mismatches between lexical assignment among speakers' native varieties contributing to lexical exceptions. Conversely, Bantu languages subject to less dialect mixing and contact-induced borrowing may be expected to retain more conservative, more semantically productive noun class systems. This sort of cross-linguistic variation is fully compatible with the fossils framework, and applying it systematically across the Bantu family would be a natural extension of the present work.

As should be clear by now, the notion of grammatical fossils is not specific to Swahili, nor to noun class semantics. Any measurable pattern in linguistic data may or may not be enforced by productive grammatical rule, and instances whose origins are better explained diachronically than synchronically appear across various domains. Estonian phonology provides a candidate fossil for consideration. Among the Finnic languages, Standard Estonian is somewhat unusual (but not unique) for its lack of productive vowel harmony (Harms, 1962; Vesik, 2023). Certain other Estonian dialects, as well as Standard Finnish, a close Finnic relative, have productive vowel harmony, which is reconstructable to their common ancestor (Kallio, 2014). In most Finnic languages, front and back vowels do not co-occur within a root, and this constraint extends productively to suffixes in the language. Despite its lack of productive harmony, the Estonian lexicon still shows a well-above chance tendency towards backness-matched vowels extractable by simple statistical means (Caplan & Kodner, 2018), and identifiable in large-scale statistical typological analysis (Šegedin & Cohen Priva, 2026). If vowel-harmony may be detected and quantified in a language as the above- or below-chance co-occurrence rates, then Estonian registers a weak but detectable signal, sitting between languages with productive harmony (e.g., Finnish, Hungarian, Turkish) and languages that have never had it (e.g., English, German). In the broader framework of the current paper, this is no great mystery: the distributional trend measurable in Estonian may be an eroding grammatical fossil from a formerly productive vowel alternation. Absent a process which systematically alters roots in the language, the tendency towards non-cooccurrence of some vowels within roots is carried forward epiphenomenally by learners.

Lastly, to take a potential syntactic candidate from English we have subject-auxiliary inversion following fronted negative expressions, as in “*Never have I seen such a thing*” and “*Rarely does she make mistakes*”. This construction has received sustained attention in the literature (e.g., Haegeman, 2000; Sobin, 2003), where it can be analyzed as a productive rule triggered obligatorily by fronted downward-entailing adjuncts governed by a negation feature. The diachronic source of the pattern is well-documented: Old and Middle English had productive verb-second word order (Kroch & Taylor, 1997), and what survives in Modern English may be the residue of constructions generated when V2 was still live. From the perspective of the fossil framework, the key question is whether the current construction reflects a genuinely productive rule or a fossilized set of lexical triggers that happen to share a semantic characterization. The triggering environments appear limited to a small, high-frequency set of negative polarity items — e.g., *never*, *rarely*, *seldom* — and other downward-entailing adverbs do not obviously inherit inversion (“**Infrequently does*

she...”, “**Uncommonly has he...*”), suggesting the triggering class may be closed (and lexically specified) rather than open. Furthermore, it sits alongside a handful of other apparently fossilized V2 constructions which such diverse triggers which do not form a natural class, such as locative and temporal adverbs (“*Here comes the sun*”), comparative adverbial phrases (“*So shocking was the news, that the room fell silent*”), and direct quotations (“*‘Baa,’ cried the sheep*”). A more targeted analysis would be required to adjudicate between various accounts, though any account proposing a productive rule of some kind would need to handle the disjoint distribution of these triggers.

Across potential cases here, what unifies these linguistic patterns is not the domain or the specific phenomenon in question, but the analytical stance required to diagnose them. In every case, the presence of a pattern in language data is insufficient grounds for attributing that pattern to synchronic grammar. A pattern can be real and measurable while being entirely unproductive, but a fundamentally historical artifact. Distinguishing such cases requires going beyond the data to ask, explicitly and rigorously, whether the pattern is one that learners would acquire as a productive rule.

The present paper has proposed and illustrated a tripartite framework for making this determination. The first step is to establish the existence of a pattern comprehensively, using as representative a dataset as possible and avoiding the selective sampling that has led prior work astray. The first step of the framework should be understood not merely as descriptive but diagnostic. We must establish that a pattern would not have occurred simply by chance before it merits either a grammatical or historical account in the first place. This is not a trivial step, as striking statistical patterns in language use may nevertheless have non-linguistic origins, Zipf’s Law (Zipf, 1936, 1949), governing frequency distributions requires no grammatical explanation whatsoever, as it is derivable from random typing processes (Miller, 1957), and is observed in a multitude of phenomena, many of which are purely natural with no human input. As “big data” approaches are equipped to unearth many tempting patterns, this first step of consideration is increasingly crucial (Caplan, Kodner, & Yang, 2020).

The second step is to test the synchronic status of the pattern directly, whether by targeted experimental paradigms such as the Wug test or a computational analysis. While the present study adopted the Tolerance Principle for this purpose, any theory which makes precise quantitative predictions about whether some aspect of language is enforced by the grammar could be employed. If the pattern fails this test, the third step is to seek a diachronic explanation: to ask whether a now-defunct rule could have generated the pattern at some earlier stage of the language, and whether the historical record supports such an account. Together, these three steps allow the researcher to move from a description of language data to strongly warranted claims about grammar.

The broader implication is simply that the evidentiary bar for grammatical claims must be higher than raw language data alone can satisfy, absent required and relevant theoretical scaffolding. Indeed, patterns of language use reflect the grammar, but they also may reflect traces of history, contact, language change, or chance. Carefully distinguishing between historical effects and grammatical causes is a prerequisite for building accurate theories of linguistic knowledge.

Acknowledgments

The authors thank Jason Kandybowicz for useful comments.

Competing Interests

The authors have no competing interests to declare.

Data availability

GitHub repository with data and analysis code to be made available pending anonymous review.

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A Appendix

A.1 word2vec Training Parameters

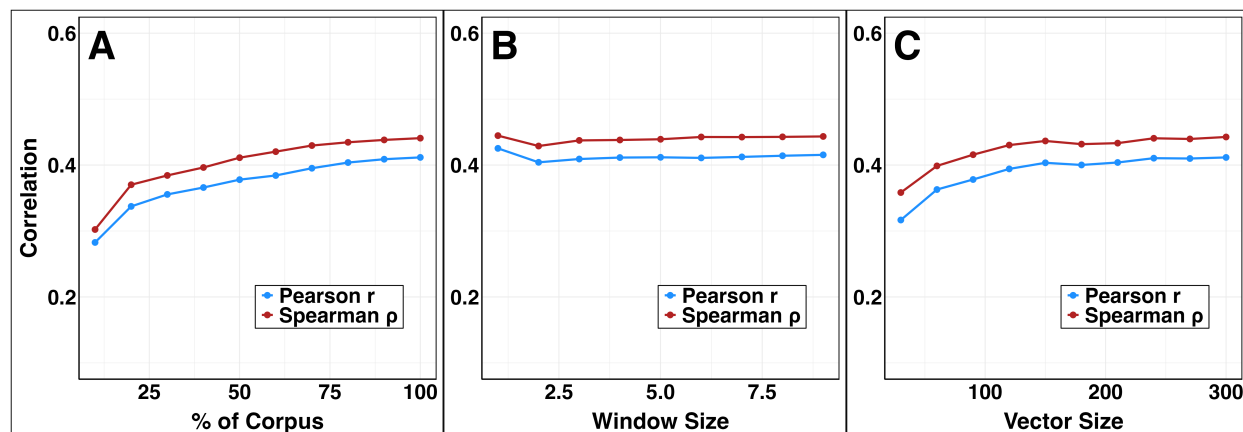


Figure A.1: Various preliminary experiments prior to word2vec training. Each measures Spearman and Pearson correlations with human similarity judgments as a proxy for embedding quality. Plot A measures embedding quality across different training set sizes. The quality plateaus as the percent of the corpus shown increases, suggesting that a larger corpus would not have increased the embedding quality. Plot B tests different context window sizes on embedding quality. These are roughly flat, suggesting that window size does not have a large impact on embedding quality. Plot C tests embedding quality across vectors of differing dimension sizes. Embedding quality plateaus around 180 dimensions. To be safe, 300-dimension vectors were elected.

A.2 Semantic Compactness Analysis

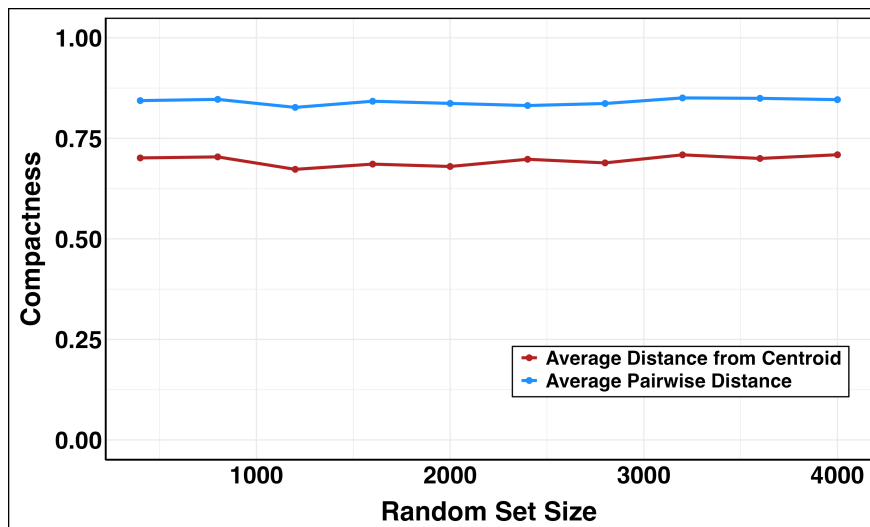


Figure A.2: Two compactness measures (average pairwise distance and average distance from centroid) for differently-sized sets of randomly selected nouns. It is important to determine whether a relationship between the number of words and their compactness exists. These results suggest that no such relationship exists, meaning it was not necessary to fix a random set’s size to the noun class set against which it is compared, e.g. in Figures 2 and A.5.

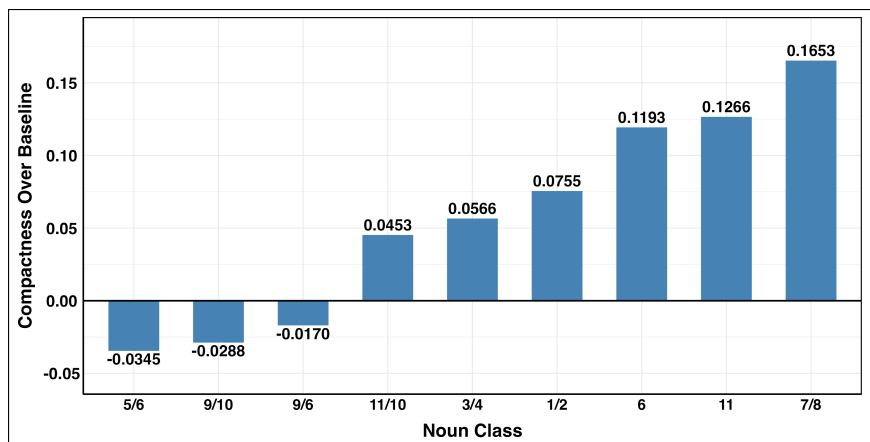


Figure A.3: Semantic compactness of noun class groups using fastText embeddings (instead of word2vec). The y-axis measures how much more compact these sets were than the random set (fixed at $y = 0$), with respect to vectors’ **average pairwise distance**. The higher the value, the more compact the sets were. The three class pairs whose compactness values are below random (5/6, 9/6, and 9/10) are the only three class pairs that lack an overt class prefix. These results suggest that fastText’s sensitivity to substrings “favored” (i.e., clustered) nouns with the same class prefix, resulting in higher compactness for those class pairs. Because the experiment is meant to target lexical semantics rather than morphology, fastText was deemed less favorable than word2vec.

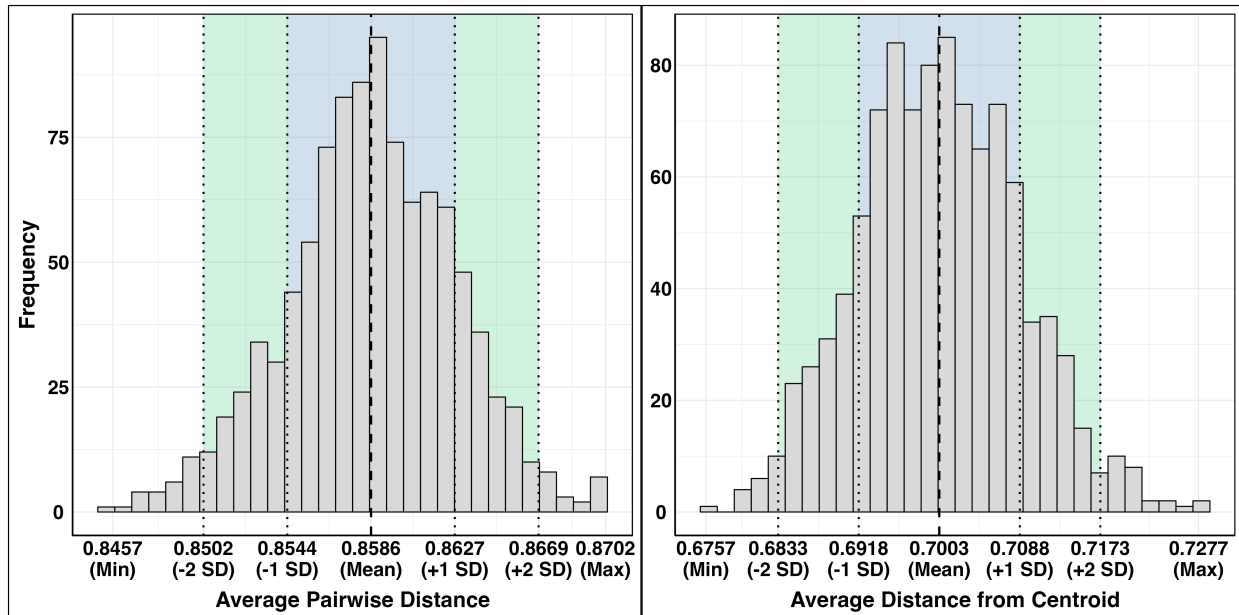


Figure A.4: Histogram of compactness values (average pairwise distances on left, average distances from centroid on right) for 1000 sets of randomly selected words across all noun class pairs. The mean value was subtracted from compactness scores to give the “Compactness Over Baseline” values in Figure 2. $\pm 1SD$ is included in the shaded blue region, and an additional SD in the green region. These outer bounds correspond to the dotted lines in Figure 2.

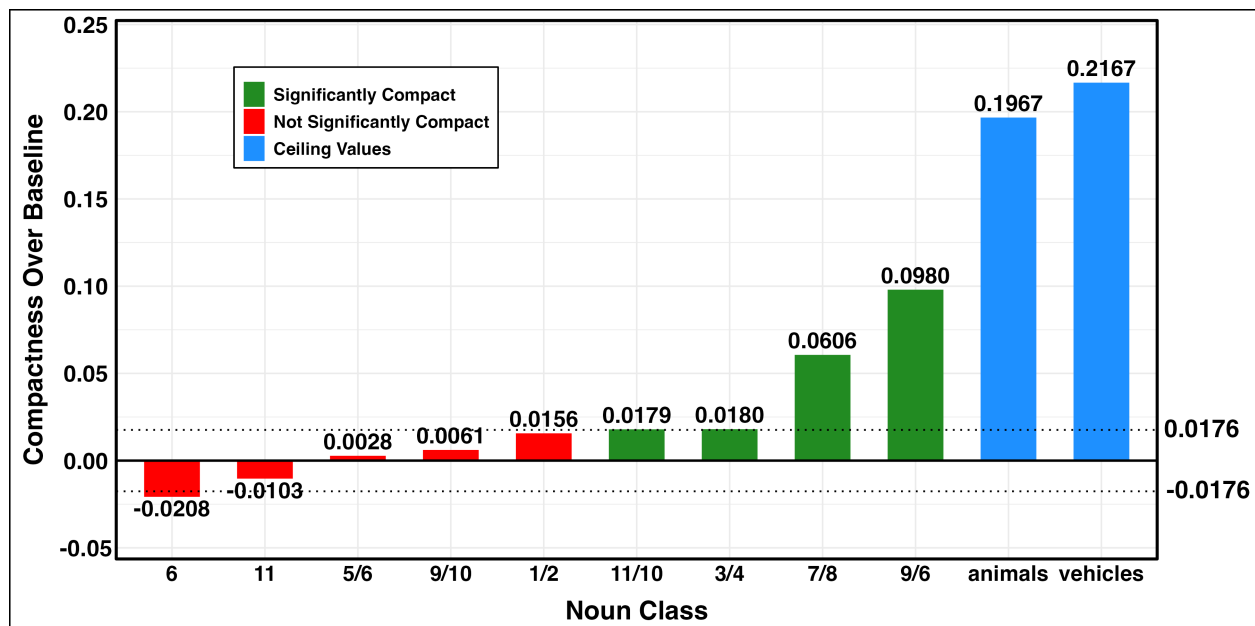


Figure A.5: Semantic compactness of noun class groups alongside two ceiling groups: animals and vehicles. The y-axis measures how much more compact these sets were than the random set (fixed at $y = 0$), with respect to vectors’ **average distance from centroid**. The higher the value, the more compact the sets were. The dotted lines represent two standard deviations above the random set’s compactness distribution.

A.3 Tolerance Principle and Loanword Analysis

Semantic Feature $\Rightarrow$	Class	N	e	Class $\Rightarrow$	Semantic Feature	N	e
HUMAN	1/2	96	27	1/2	HUMAN	70	1
HUMAN	1/2*	96	0	1/2	HUMAN*	97	1
TREE	3/4	1	0	3/4	TREE	49	48
DISPERSIVE MASS	3/4	1	1	3/4	DISPERSIVE MASS	49	49
EXTENDED	3/4	9	5	3/4	EXTENDED	49	45
SPREAD	3/4	4	4	3/4	SPREAD	49	49
COHESIVE MASS	5/6	5	2	5/6	COHESIVE MASS	79	76
COHESIVE MASS	11, 11/10	5	5	11, 11/10	COHESIVE MASS	56	56
AUGMENTATIVE	5/6	1	1	5/6	AUGMENTATIVE	79	79
PLANT PART	5/6	1	0	5/6	PLANT PART	79	78
FRUIT	5/6	1	0	5/6	FRUIT	79	78
ROUND	5/6	2	1	5/6	ROUND	79	78
ARTEFACT	7/8	9	6	7/8	ARTEFACT	35	32
PEJORATIVE	7/8	1	1	7/8	PEJORATIVE	35	35
PLANT	7/8	2	2	7/8	PLANT	35	35
MANNER	7/8	0	0	7/8	MANNER	35	35
ANIMAL	9/10	4	2	9/10	ANIMAL	211	209
NARROW	11, 11/10	8	8	11, 11/10	NARROW	56	56
WAVY	11, 11/10	1	1	11, 11/10	WAVY	56	56
ABSTRACT	11, 11/10	229	181	11, 11/10	ABSTRACT	56	8
TREE + DISPERSIVE MASS + EXTENDED + SPREAD	3/4	15	10	3/4	TREE + DISPERSIVE MASS + EXTENDED + SPREAD	49	45
COHESIVE MASS + AUGMENTATIVE + PLANT PART + FRUIT + ROUND	5/6	10	4	5/6	COHESIVE MASS + AUGMENTATIVE + PLANT PART + FRUIT + ROUND	79	73
ARTEFACT + PEJORATIVE + PLANT + MANNER	7/8	12	9	7/8	ARTEFACT + PEJORATIVE + PLANT + MANNER	35	32
NARROW + WAVY + COHESIVE MASS + ABSTRACT	11, 11/10	243	195	11, 11/10	NARROW + WAVY + COHESIVE MASS + ABSTRACT	56	8

Table A.1: Tolerance Principle analysis of rules between proposed semantic features and noun classes. Left-side columns test the necessary condition (whether feature predicts class), where N is the number of nouns bearing the feature and e is the number of those nouns *not* assigned to that class. Right-side columns test the sufficient condition (whether class predicts semantics), where N is the number of nouns in the class and e is the number not expressing the listed feature. Green highlighted rows indicate a productive pattern under the TP ($e < \theta$). Following Yang (2016), small samples ($N < 5$) are excluded. Asterisks denote cases where agreement class diverges from noun class prefix: HUMAN nouns without a Class 1/2 prefix typically trigger Class 1/2 agreement regardless.